

Background

- Rheumatology medications are complex, high-risk therapies that may benefit from management with a team-based approach including a pharmacist¹
- A specialty pharmacy within a health system provides support for medication monitoring, patient education, prior authorizations, financial assistance, dispensing and delivery
- Pharmacists integrated into rheumatology clinics as an extension of the health system specialty pharmacy can collaborate with clinicians to mitigate treatment barriers including medication safety and enhance the continuity of care in the ambulatory setting ^{2,3}

Objectives

- Develop an integrated model for ambulatory care pharmacy within rheumatology clinics to facilitate patient education, specialty medication monitoring, adherence, and improve clinical continuity (determined by the number of prescriptions sent to health system pharmacy) with the integrated health system specialty pharmacy

Methods

Research	- Background research, stakeholder analysis completed to identify a rheumatology clinic site to integrate pharmacist services - Identify clinics with high utilization of self-administered specialty medications
Develop	- Specialty collaborative practice agreement - Pharmacist visit workflow
Integrate	- Pharmacist recruitment, credentialing - Administrative space and exam rooms allocated for the pharmacist - Clinic go-live
Assess	Primary outcomes: - Clinical continuity - Number of referred patients Secondary outcomes - Medication adherence - Proportion of Days Covered (PDC) - Disease activity - Routine Assessment of Patient Index Data 3 (RAPID3)

Results

Figure 1: Collaborative Practice Agreement: Rheumatology Appendix

Disease States	Therapeutic Classes	Actions	Labs
- Inflammatory Arthritis disorders - Ankylosing Spondylitis - Systemic Lupus Erythematosus and associated disorders	- Corticosteroids - Janus Kinase Inhibitors - Disease-Modifying Antirheumatic Drugs - Biologics¹ - Aminoquinoline - Mycophenolate	- Initiate - Refill - Modify - Discontinue	Baseline infection screenings - Tuberculosis testing - Hepatitis B and C serology - HIV testing Disease state progression - C-reactive protein - Sedimentation rate Other disease state impact - Hemoglobin A₁C - Fasting glucose - Lipid panel Medication monitoring - Complete Blood Count with differential - Absolute neutrophil count - Comprehensive metabolic panel - Hemoglobin - Hematocrit - Reticulocyte count - Platelet count

1) Example biologics include but are not limited to tumor necrosis factor (TNF) inhibitors, interleukin (IL) 1 inhibitors, IL-6 inhibitors, phosphodiesterase (PDE)-4 inhibitors, IL-12/23 inhibitors, IL-17a inhibitors and monoclonal antibody

- The first patient was referred to the integrated pharmacist in November 2020 and the collaborative practice agreement was approved in May 2021
- In the first six months, the project expanded to two additional pharmacists serving three other clinics
- Primary outcome data was collected for the three clinics
- Target clinical continuity is 64% by end of the first year
- Secondary outcome data are still being collected

Figure 2: Overall Proportion of Days Covered

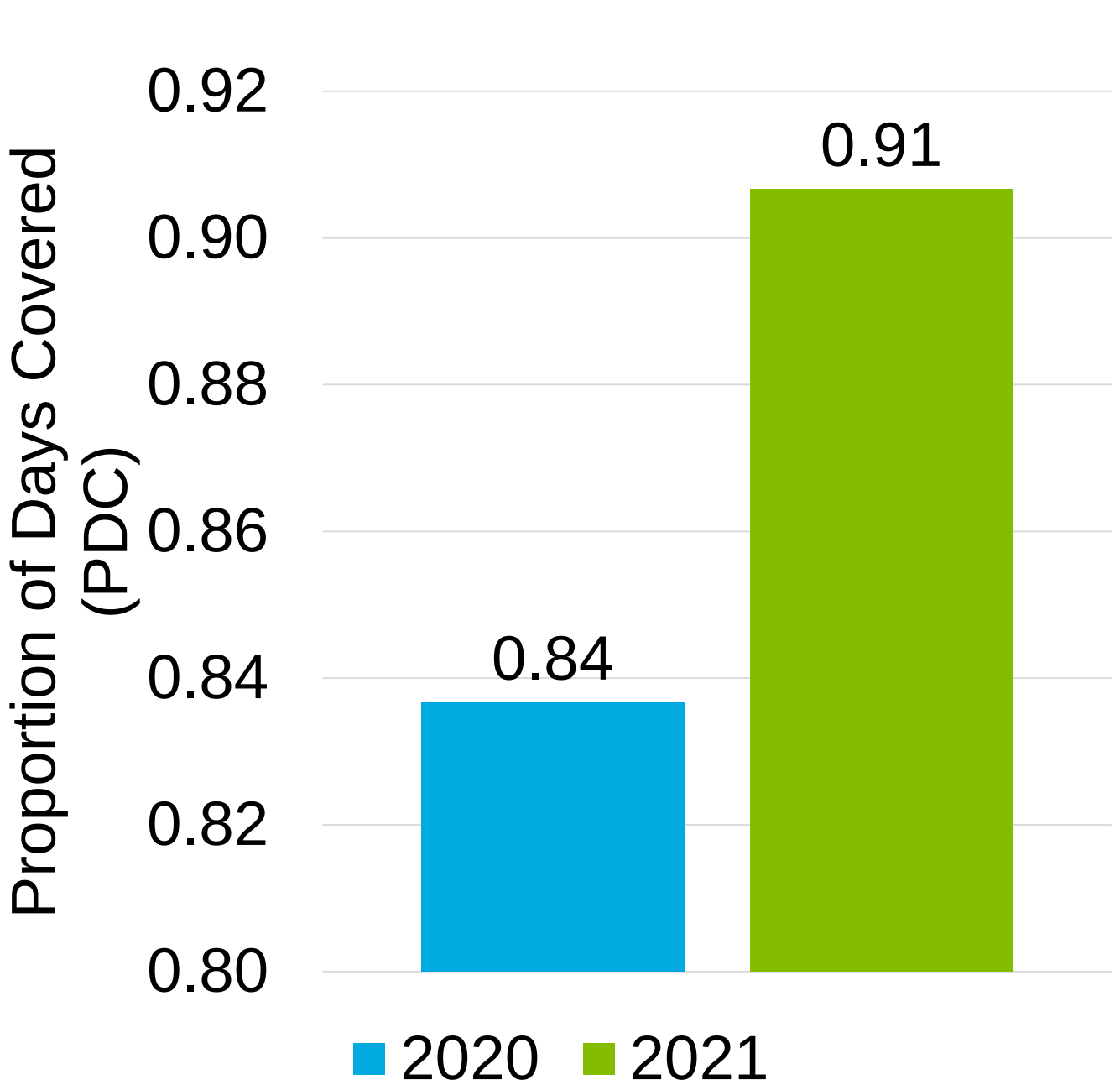
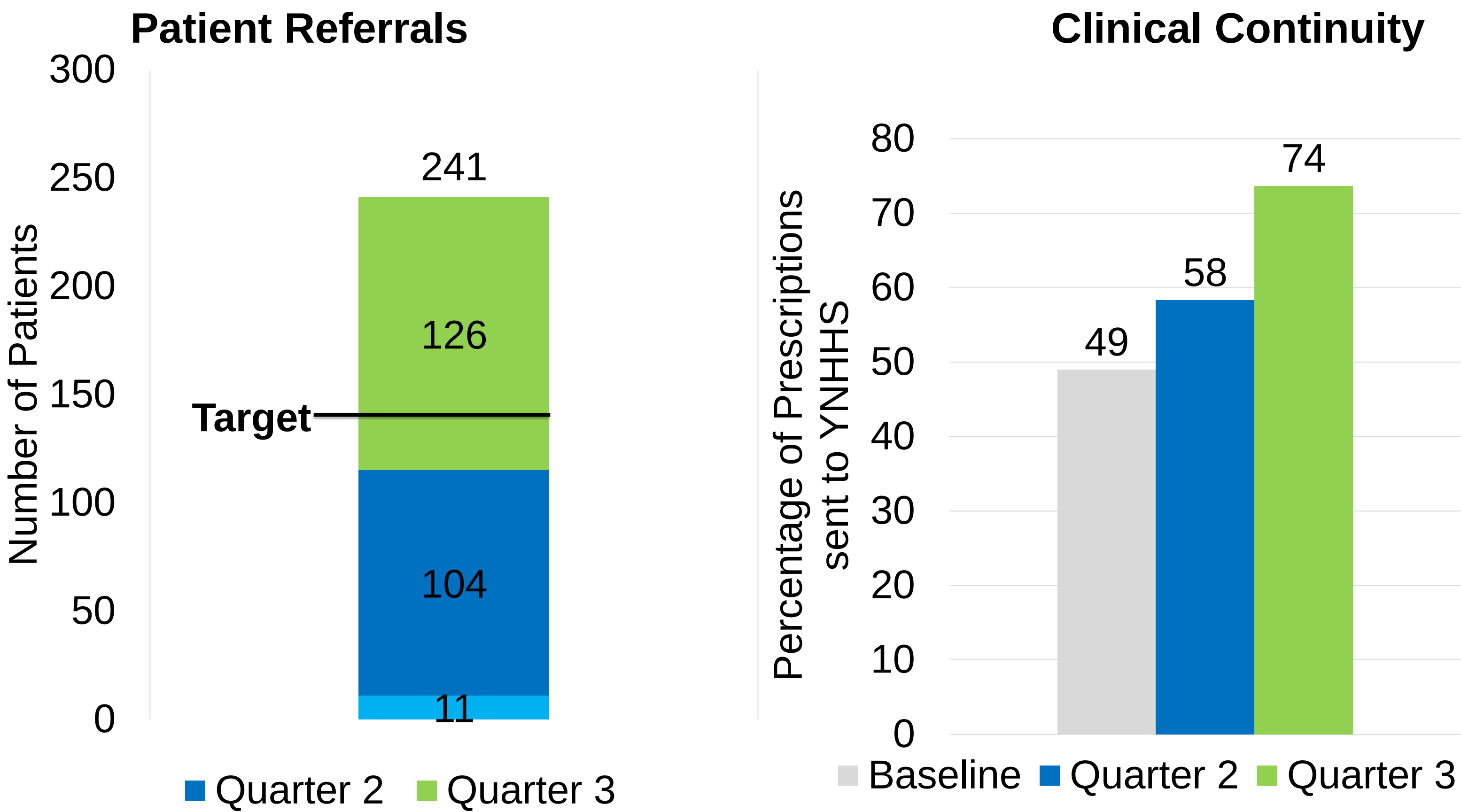


Figure 3: Clinical Continuity and Referrals



Discussion

- Within six months of implementation, the number of pharmacist referrals generated exceeded the target goal
- Integration of ambulatory care pharmacists into rheumatology clinics increased clinical continuity of specialty prescriptions
- Achievements demonstrated by the pilot clinic expedited the expansion of pharmacy services to additional clinic sites
- Medication adherence remained high, with a slight increase from baseline
- Integrated pharmacists positively impacted patient care, including medication education, affordability, and safety
- Limitations included ability to extrapolate data, including RAPID3 assessment scores and lab adherence

Conclusions

- A model for integrating ambulatory care pharmacists within target rheumatology clinics was developed and implemented
- Implementation exceeded target for referred patients by 151%, resulted in a 25% increase in clinical continuity, and an 8% increase in patient adherence, as measured by PDC

Future Directions

- Expand integrated pharmacist services within the health system to all rheumatology departments and to other specialties such as dermatology
- Optimize data reporting to collect lab monitoring adherence efficiently

References: 1) Ferguson, P.J., Lynton, J., Resman-Targoff, B.H. (2020 Nov) Advantages of Embedding a Specialty Pharmacist in a Rheumatology Clinic. *The Rheumatologist*. Available from <https://www.the-rheumatologist.org>. 2) Farrell J, Shapiro LS, Miller M. (2017). Clinical Pharmacist As Part of the Interprofessional Team Improves Quality of Care in Patients with Rheumatic Disease [abstract]. *Arthritis Rheumatol*. 2017; 69 (suppl 10). Available from <https://acrabstracts.org/>. 3) DeBenedette, V. (2019 Sept) Incorporating the pharmacist into the rheumatology practice. Rheumatology Network. Available from www.rheumatologynetwork.com. The authors of this presentation have nothing to disclose concerning possible financial or personal relationships with commercial entities that may have a direct or indirect interest in the subject matter of this presentation. National Association of Specialty Pharmacy Conference. Virtual Viewing September 24, 2021